



Part 2:

Reimagining Neurorehabilitation: Enabling scalable, affordable and sustainable care through intelligent BCI-controlled exoskeletons

Prof. Girijesh Prasad

Director of NI Functional Brain Mapping Facility

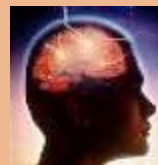
Intelligent Systems Research Centre,

School of Computing, Engineering & Intelligent Systems

Ulster University, Derry~Londonderry, NI, UK.

g.prasad@ulster.ac.uk

ulster.ac.uk



Talk roadmap

Designed for a 30-minute workshop talk with adoption-oriented transitions.

1

Problem context

Why stroke rehabilitation needs scalable AI-enabled care models

2

System architecture

From BCI signal acquisition to embodied exoskeleton feedback

3

Trustworthy adaptation

Managing non-stationarity, fatigue and human-in-the-loop safety

4

Clinical evidence

Pilot results, recovery metrics and prognostic signals

5

LMIC adoption lens

Decentralisation, patient communication and workflow integration

Why this matters for LMIC healthcare

20M

annual stroke cases worldwide
in the source deck

~65%

survive stroke

~50%

live with hand disability after
stroke

6 weeks

pilot therapy schedule in the
reported protocol

Care-system pressure

Rehabilitation demand extends far beyond acute care and often outlasts specialist availability.

Adoption implication

AI adoption is credible only if systems are affordable, explainable and usable outside tertiary centres.

Neurorehabilitation is a strong case study for AI adoption because it links patient communication, workflow redesign and equitable access.

Framing innovation around SDG3 shifts attention from proof-of-concept to scalable public benefit.

A trustworthy AI framing for neurorehabilitation

Trustworthy

Selective adaptation, clinician oversight and safety-aware fallback logic.

Decentralised

Potential for home-operable and app-connected pathways that distribute care closer to the patient.

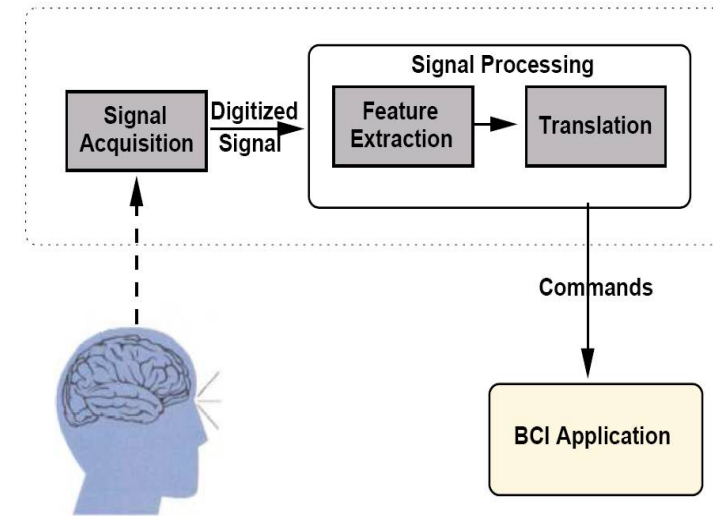
Workflow-aware

Supports therapists, patients and caregivers through monitoring, alerts and interpretable escalation.

This framing aligns the science with workshop questions on trustworthy, decentralised AI in healthcare.

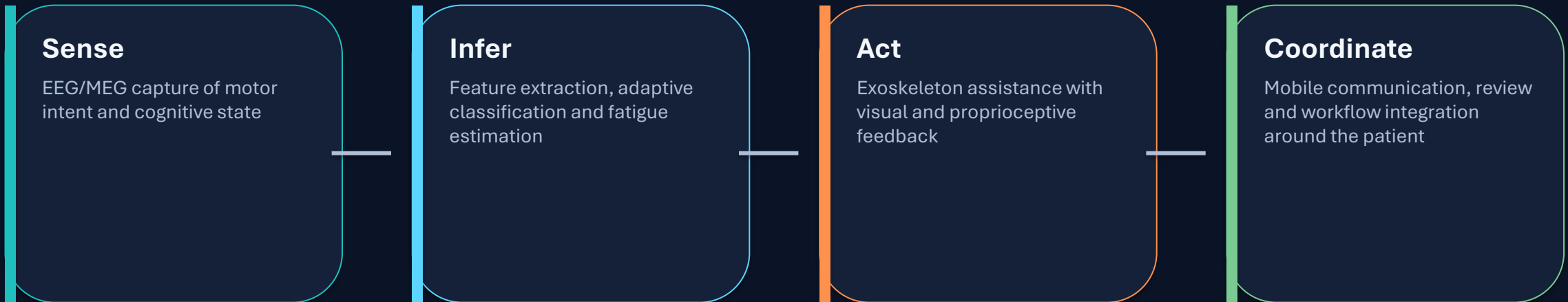
Integration of BCI and robotic exoskeleton : A way forward for neurorehabilitation

- Sustained therapy by physiotherapists is limited & prohibitively expensive;
- Robots administer repetitive exercises but may not be sufficiently engaging for neuroplasticity to occur due to:
 - lack of interactive and engaging user interface.
- A symbiotic fusion of BCI and robot can facilitate:
 - Engaging and natural user interface;
 - Capturing of patient's attention while allowing intense repetition of therapeutic task, causing neuroplasticity;
 - Reliable quantification of patient's performance and mental fatigue;
 - Estimation of potential motor recovery prospects - prognosis.



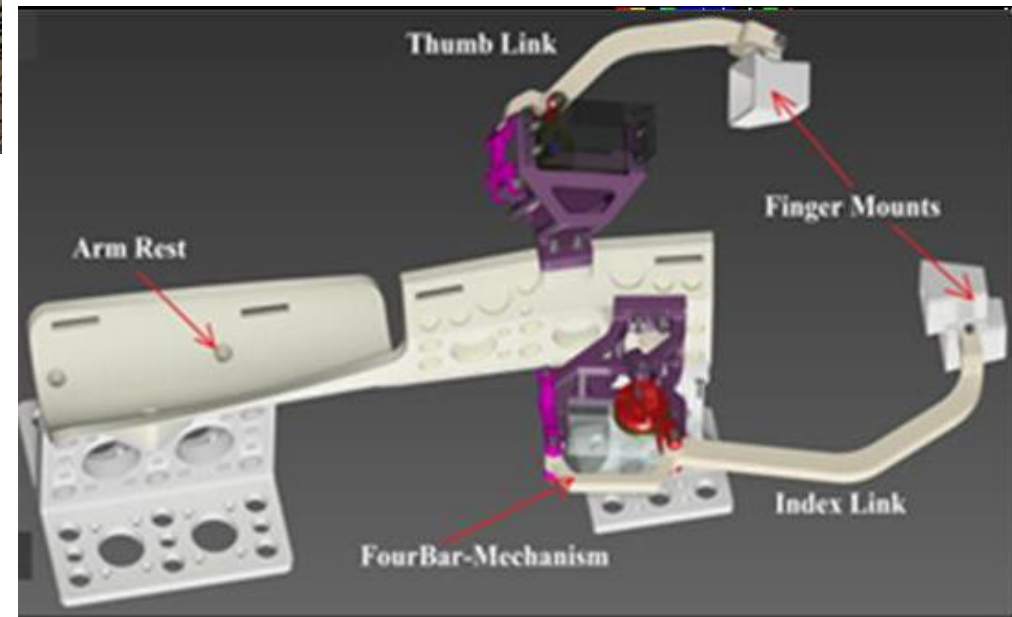
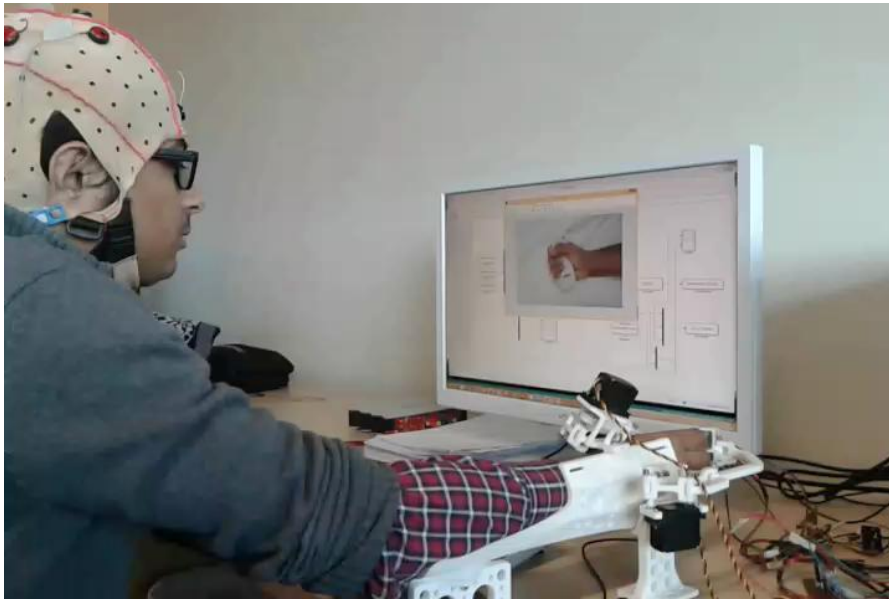
MIT-MANUS (Krebs et al., 2004)

BCI-exoskeleton pipeline for adoption



The coordination layer is what transforms a lab pipeline into an adoption pathway.

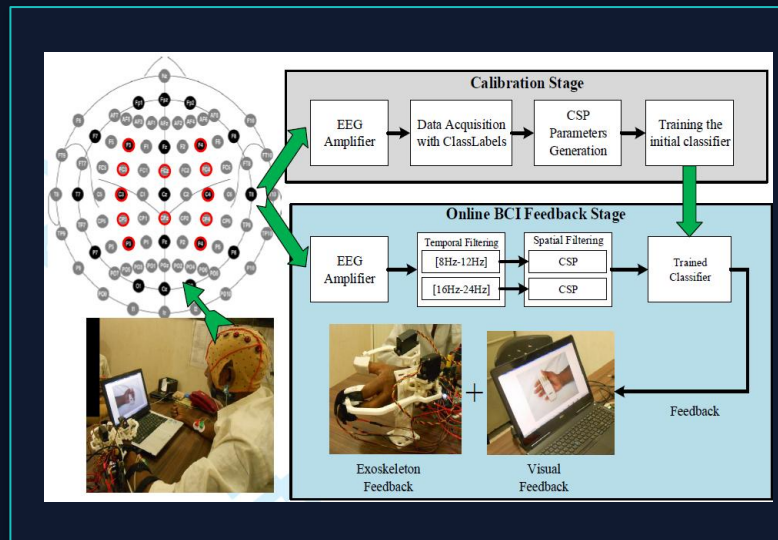
A Participant using the Neuro-rehabilitation System



[Chowdhury et al. (2018), *IEEE T Cognitive and Developmental Systems*]

EEG BCI pathway: practical intelligence close to care

CSP and cortico-muscular coupling / EEG-EMG CBPT features provide a clinically relevant control signal. SVM-based decoding is attractive because it is comparatively interpretable and operationally lightweight. The translational challenge is maintaining stable performance across sessions rather than maximising one-off accuracy.



Translational emphasis

Operational simplicity and stability matter as much as accuracy.

Why it suits decentralisation

Lightweight local inference can be more feasible than infrastructure-heavy centralised pipelines.

Clinical value

Embodied feedback makes the AI legible to patients and therapists through action, not only scores.

TRUST

Why adaptive AI matters in the clinic

Risk

Brain signals drift within and across sessions, making static models brittle.

Trust value

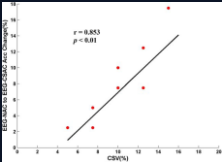
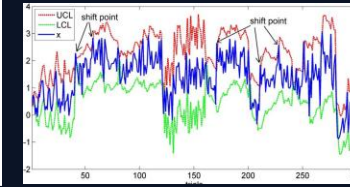
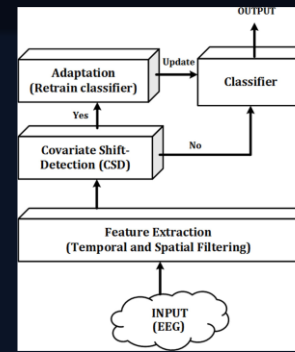
Selective updating is easier to justify than opaque continuous retraining.

Method

Covariate-shift detection triggers adaptation only when needed.

Deployment value

Lower compute burden matters for distributed and lower-resource care settings.



Pilot clinical protocol

Selection

Stroke participants with partial or full finger-motion disability

Eligibility

At least 6 months post-stroke; MMSE above 21/30

Intervention

30 min assist-as-needed physical practice + 30 min BCI mental practice

Schedule

2-3 sessions per week for 6 weeks

EVIDENCE

Pilot outcomes that matter for adoption

Behavioural recovery

All participants achieved MCID in ARAT

Neurophysiology

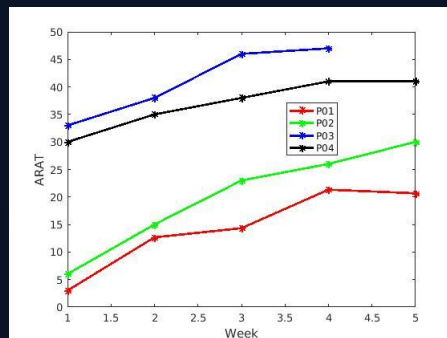
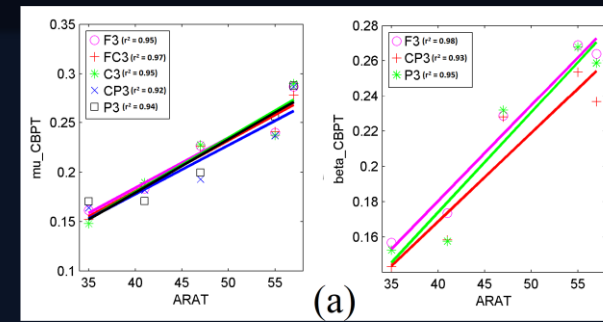
Mu and beta ERD/ERS shifts increased from first to last session

Prognosis

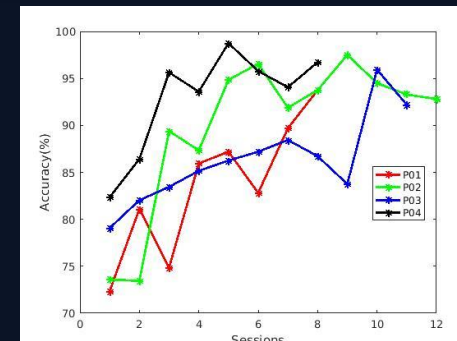
CBPT patterns tracked motor recovery potential

Model performance

BCI accuracy improved across sessions for all four participants



[Chowdhury et al.
(2020), IEEE
Access]

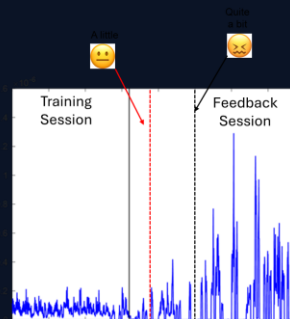
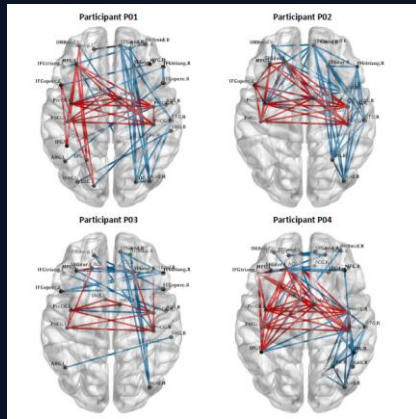


MEG extends the framework beyond control

MEG supports richer markers for connectivity, channel selection and mental fatigue.

Passive monitoring can indicate whether a participant is ready to continue, pause or retune the task.

That broadens AI from device control to clinically meaningful session management.



Control

Intent decoding

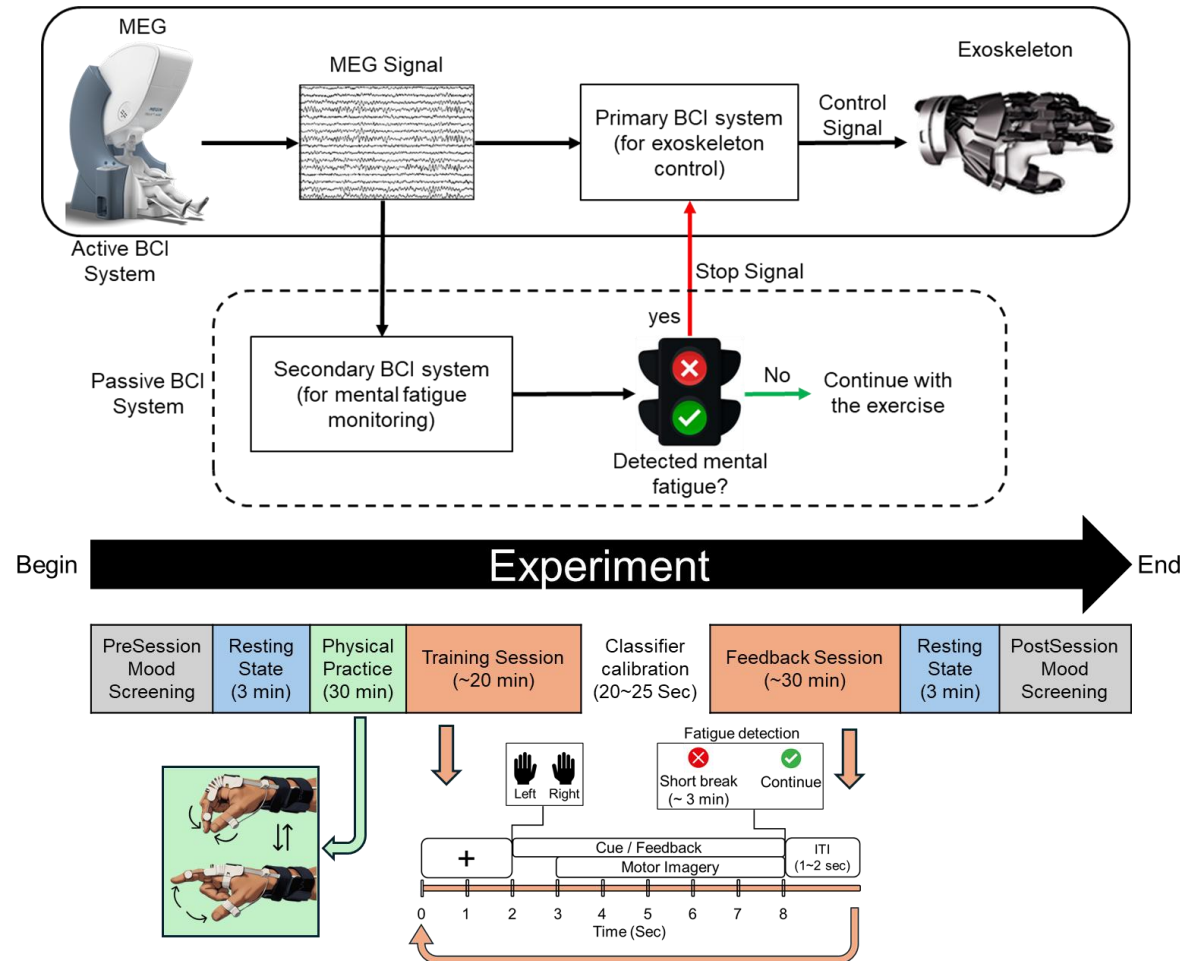
Monitor

Fatigue and readiness

Manage

Pause, continue or adapt

Advancing MEG-based BCI Supported Neurorehabilitation



Patient communication and clinical workflows

Patient communication

App-based prompts can translate model outputs into understandable cues, progress updates and fatigue alerts.

Clinical workflows

Therapists can supervise by exception rather than constant in-person monitoring.

Decentralised care

Monitoring and escalation can be distributed across home, caregiver and clinic contexts.

LMIC adoption considerations

Design for intermittent connectivity and limited specialist workforce from the outset.
Use modular architectures so sensing, inference and communication can sit across device, phone and clinic.
Prioritise multilingual patient interfaces, confidence-aware alerts and safe fallback modes.

Confidence-aware alerts and safe fallback behaviour are essential where oversight bandwidth is limited.

Scientific and translational futures

Research need

Robust biomarkers for prognosis and mental fatigue

Evidence need

Larger randomised controlled studies across more diverse groups

Systems need

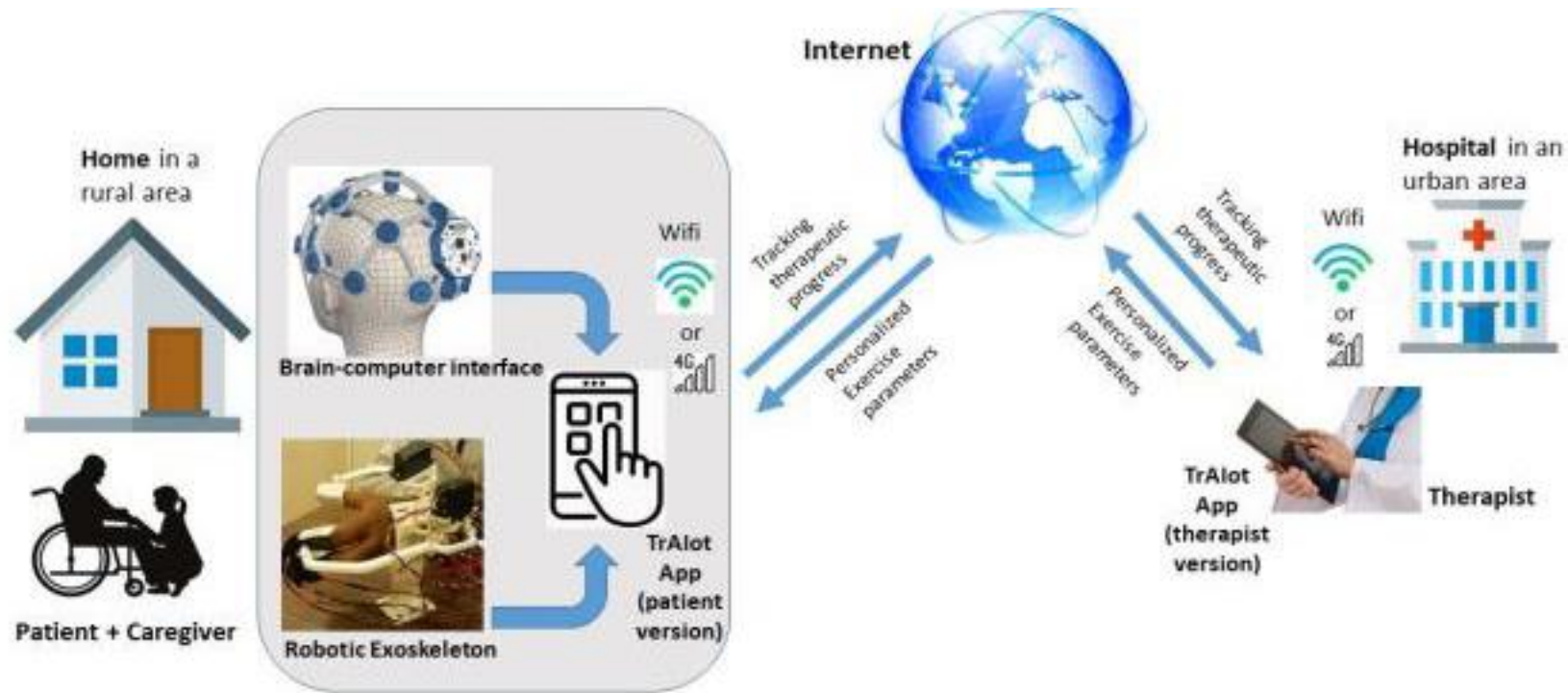
Self-adapting home-operable IoT neurorehabilitation platforms

Community asset

Open MEG BCI datasets to widen participation and benchmarking

Work-in-Progress

- Developing a **self-adapting home-operable IoT** system monitored and controlled through a mobile NeuroRehab app.



Take-home message

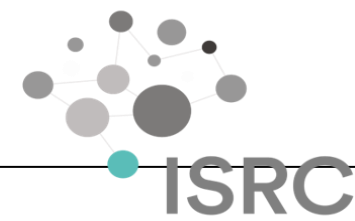
The strongest case for rehabilitation AI in LMICs is trustworthy augmentation of scarce clinical capacity through decentralised, human-centred systems.

Further...

- For wider research community participation, MEG BCI data is made available in:
 - D. Rathee, H. Raza, S. Roy, G. Prasad, “A magnetoencephalography dataset for motor and cognitive imagery-based brain-computer interface,” **Scientific Data-Nature**, vol. 8, no. 1, p. 120, 12, 2021.

[10.6084/m9.figshare.c.5101544.v1](https://doi.org/10.6084/m9.figshare.c.5101544.v1)

Acknowledgments...



Ulster University, UK :

- S McDonough (Prof. Health and Rehabilitation)
- D H Coyle (Prof. Neurotechnology)
- K F Wong-Lin (Professor, SCEIS)
- S. Bhattacharyya (Lecturer, SCEIS)
- K McCreadie (Lecturer, SCEIS)
- R P Beveridge (Tech Services Co-Ordinator)
- Kaniska Samanta (Former PhD researcher)
- S. Roy (Former PhD researcher)
- Y.K. Meena (Former PhD researcher)
- D. Rathee (Former PhD researcher)
- H. Raza (Former PhD researcher)
- P. Gaur (Former PhD researcher)
- V. Youssofzadeh (Former PhD researcher)

Altnagelvin Hospital, Derry, UK:

- Annmarie Kelly (Occupational Therapist)

IIT Kanpur India:

- A. Dutta (Prof. Mechanical Eng.)
- B. Bhushan (Prof. H & S Sci.)
- KS Venkatesh (Prof. Electrical Eng.)
- P. Yadav (PhD researcher)
- A. Choudhury (Former PhD researcher)
- S.S. Nishad (PhD researcher)

Regency Hospital, Kanpur, India:

- A. Bajpai (DM)
- N. Pandey (DM, Neurology)
- A. A. Hashmi (DM, Neurology)
- A.K. Uttam (DM, Neurology)

➤ **Funding support from:**

- UK India Education and Research Initiative (UKIERI) phases 1, 2 and 3
- InvestNI & Ulster University for NIFBM Facility (1303/101154803)
- MRC (MC_PC_MR/X013243/1)
- UKRI Strength in Places Fund for Smart Nano NI

